7.1 Right Triangles

You can use the triangle congruence theorems from the last chapter to develop theorems specifically for right triangles. Let's briefly review the terminology needed to study these right triangles.

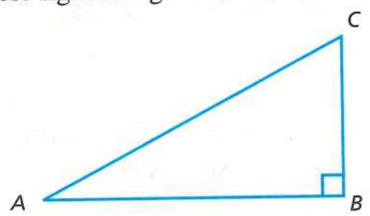


Figure 7.1

Why are the right triangle windows isosceles?

In right $\triangle ABC$, $\angle B$ is the right angle. The side opposite the right angle, $\overline{AC}$, is the *hypotenuse*. The other sides of the triangle, $\overline{AB}$ and $\overline{BC}$, are the *legs* of the right triangle.

The most important congruence theorem for right triangles is HL (Hypotenuse-Leg). However, its proof requires auxiliary lines. Once proved, it will provide an important way to prove two right triangles congruent.

Theorem 7.1

HL Congruence Theorem. **If the hypotenuse and a leg of one right triangle are congruent to the hypotenuse and corresponding leg of another right triangle, then the two triangles are congruent.**

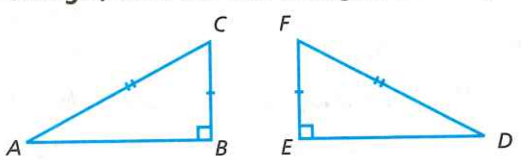


Figure 7.2

Given: $\overline{AC} \cong \overline{DF}$; $\overline{BC} \cong \overline{EF}$; $\angle B$ and $\angle E$ are right angles
Prove: $\triangle ABC \cong \triangle DEF$

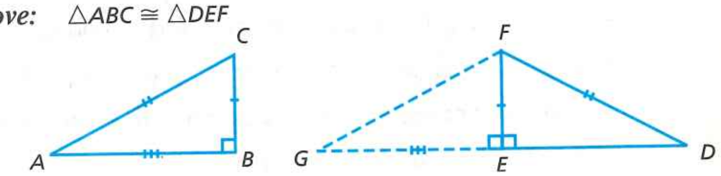


Figure 7.3

STATEMENTS	REASONS
1. $\overline{AC} \cong \overline{DF}$; $\overline{BC} \cong \overline{EF}$; $\angle B$ and $\angle E$ are right angles	1. Given
2. There is a point, G, on the ray opposite $\overrightarrow{ED}$ such that $\overline{EG} \cong \overline{AB}$; draw $\overline{FG}$	2. Auxiliary lines
3. $\angle FEG$ and $\angle FED$ form a linear pair	3. Definition of linear pair
4. $\angle FEG$ is a right angle	4. If one of the angles in a linear pair is a right angle, the other is also a right angle
5. $\angle B \cong \angle FEG$	5. All right angles are congruent
6. $\triangle ABC \cong \triangle GEF$	6. SAS
7. $\overline{AC} \cong \overline{FG}$	7. Definition of congruent triangles
8. $\overline{DF} \cong \overline{FG}$	8. Transitive property of congruent segments
9. $\triangle GDF$ is isosceles	9. Definition of isosceles triangle
10. $\angle FGE \cong \angle FDE$	10. Isosceles Triangle Theorem
11. $\triangle GEF \cong \triangle DEF$	11. SAA
12. $\triangle ABC \cong \triangle DEF$	12. Transitive property of congruent triangles

The other ways to prove right triangles congruent are LL (Leg-Leg), LA (Leg-Angle), and HA (Hypotenuse-Angle). These are easier to prove because they are special cases of theorems in the previous chapter. The only special feature is that the triangles are right, so the right angles are congruent. Study the proof of LL so that you will be able to do the others. HA is exercise 10. LA has two cases (exercises 11-12), depending on whether the leg is opposite or adjacent to the angle.

Theorem 7.2

LL Congruence Theorem. **If the two legs of one right triangle are congruent to the two legs of another right triangle, then the two triangles are congruent.**

Given: $\triangle HIJ$ and $\triangle MLN$ are right triangles; $\overline{HI} \cong \overline{ML}$; $\overline{JI} \cong \overline{NL}$
Prove: $\triangle HIJ \cong \triangle MLN$

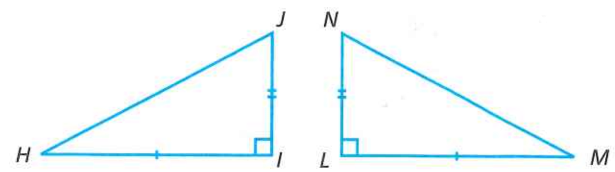


Figure 7.4

STATEMENTS	REASONS
1. $\triangle HIJ$ and $\triangle MLN$ are right triangles; $\overline{HI} \cong \overline{ML}$; $\overline{JI} \cong \overline{NL}$	1. Given
2. $\angle I$ and $\angle L$ are right angles	2. Definition of right triangle
3. $\angle I \cong \angle L$	3. All right angles are congruent
4. $\triangle HIJ \cong \triangle MLN$	4. SAS
5. If two legs of one right triangle are congruent to the two legs of another right triangle, then the triangles are congruent	5. Law of Deduction

Theorem 7.3
HA Congruence Theorem. **If the hypotenuse and an acute angle of one right triangle are congruent to the hypotenuse and corresponding acute angle of another right triangle, then the two triangles are congruent.**

Theorem 7.4
LA Congruence Theorem. **If a leg and one of the acute angles of a right triangle are congruent to the corresponding leg and acute angle of another right triangle, then the two triangles are congruent.**

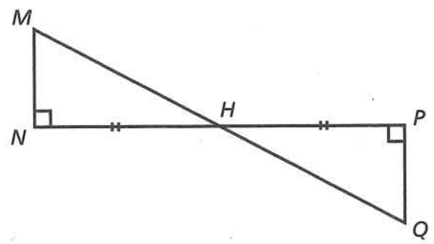
▶ A. Exercises

1. State the four methods of proving that two right triangles are congruent. Draw a picture showing each case and mark the triangles accordingly.

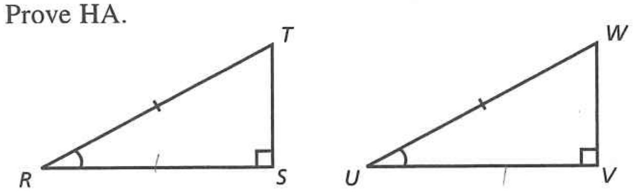
Identify any triangle congruence theorem or postulate corresponding to each right triangle congruence theorem below. (Remember that the right angles are congruent.)

2. LL
3. LA, adjacent case
4. LA, opposite case
5. HL
6. HA
7. Which theorem for right triangle congruence does not correspond to another congruence postulate or theorem?

8. According to exercise 7, which condition is valid for right triangles that is not otherwise valid?
9. Use the diagram to state a triangle congruence. Which right triangle theorem justifies your statement?



10. Prove HA.

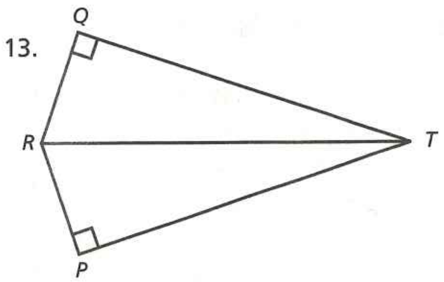


▶ B. Exercises

Use the same diagram as in exercise 10 for the proofs in exercises 11-12 (the two cases of the LA Congruence Theorem).

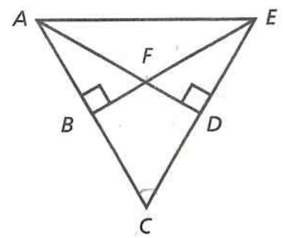
11. LA (opposite case)
    Given: $\triangle RST$ and $\triangle UVW$ are right triangles; $\overline{RS} \cong \overline{UV}$; $\angle T \cong \angle W$
    Prove: $\triangle RST \cong \triangle UVW$
12. LA (adjacent case)
    Given: $\triangle RST$ and $\triangle UVW$ are right triangles; $\overline{RS} \cong \overline{UV}$; $\angle R \cong \angle U$
    Prove: $\triangle RST \cong \triangle UVW$

Use the following diagram to prove exercise 13.

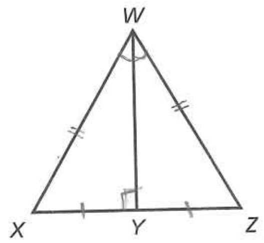


13. *Given:* $\angle P$ and $\angle Q$ are right angles; $\overline{PR} \cong \overline{QR}$
    Prove: $\overline{PT} \cong \overline{QT}$

14. Use the diagram for this proof.
    Given: $\triangle EBC$ and $\triangle ADC$ are right triangles; $\overline{EB} \cong \overline{AD}$
    Prove: $\overline{DC} \cong \overline{BC}$



Use the following diagram to prove exercises 15-19.



15. *Given:* $\overline{WY} \perp \overline{XZ}$; $\angle X \cong \angle Z$
    Prove: $\triangle XYW \cong \triangle ZYW$
16. *Given:* $\angle XYW$ and $\angle ZYW$ are right angles; $\overline{XW} \cong \overline{ZW}$
    Prove: $\triangle XYW \cong \triangle ZYW$
17. *Given:* $\angle XYW \cong \angle ZYW$; $\overline{XY} \cong \overline{ZY}$
    Prove: $\triangle XYW \cong \triangle ZYW$
18. *Given:* $\triangle XYW$ and $\triangle ZYW$ are right triangles; $\angle XWY \cong \angle ZWY$
    Prove: Y is the midpoint of $\overline{XZ}$
19. *Given:* $\triangle XYW$ and $\triangle ZYW$ are right triangles; $\overline{XY} \cong \overline{ZY}$
    Prove: $\overline{XW} \cong \overline{ZW}$

▶ C. Exercises

Use the diagram above to prove the following.

20. *Given:* $\overleftrightarrow{WY}$ bisects $\angle XWZ$; $\overleftrightarrow{WY} \perp \overleftrightarrow{XZ}$
    Prove: $\triangle XZW$ is an isosceles triangle

▶ Dominion Thru Math

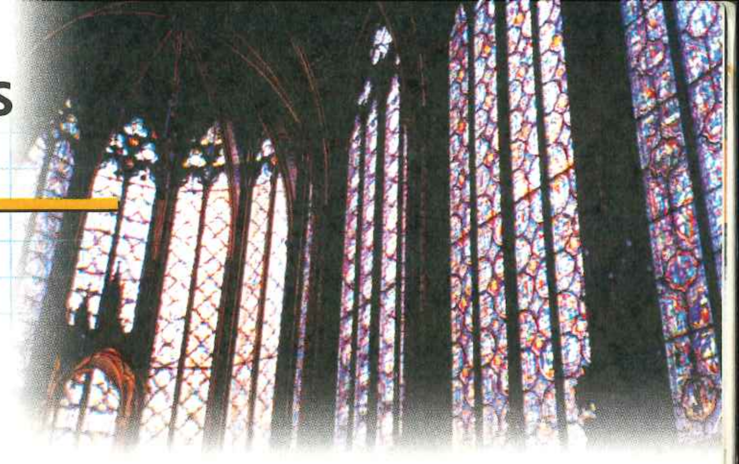
21. In miniature golf, the tee and the hole are the same distance from a wall. If you imagine the hole as if reflected in a mirror on the wall and aim at the image, the ball will bounce off the mirror into the hole. Which right triangle congruence theorem assures success by demonstrating congruent triangles?

Cumulative Review

Give the measure of the angle(s) formed by

22. two opposite rays.
23. perpendicular lines.
24. an equiangular triangle.
25. the bisector of a right angle.
26. Which symbol does not represent a set?
    $\angle ABC$, $\triangle ABC$, $A\text{-}B\text{-}C$, $\{A, B, C\}$

7.2 Perpendiculars and Bisectors



Each stained glass window of this Paris cathedral includes a bisector.

Besides right triangles, perpendicular lines are also used in several other ways. This section will present some important properties of perpendicular lines.

Theorem 7.5

Any point lies on the perpendicular bisector of a segment if and only if it is equidistant from the two endpoints.

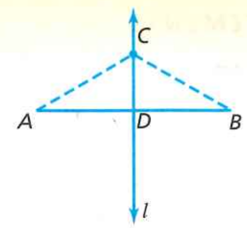


Figure 7.5

Part One (*only if*)

Given: l is the perpendicular bisector of $\overline{AB}$ at D; C is a point on l

Draw: $\overline{AC}$ and $\overline{BC}$

Prove: $AC = BC$

STATEMENTS	REASONS
1. $\overline{AB} \perp l$ at D; C is a point on l	1. Given
2. Draw $\overline{AC}$ and $\overline{CB}$	2. Auxiliary lines (Line Postulate)
3. $\angle CDA$ and $\angle CDB$ are right angles	3. Definition of perpendicular
4. $\triangle CDA$ and $\triangle CDB$ are right triangles	4. Definition of right triangle
5. $AD = DB$	5. Definition of segment bisector
6. $\overline{AD} \cong \overline{DB}$	6. Definition of congruent segments
7. $\overline{CD} \cong \overline{CD}$	7. Reflexive property of congruent segments

Continued ▶

8. $\triangle CDA \cong \triangle CDB$	**8.** LL (or SAS)
9. $\overline{AC} \cong \overline{BC}$	**9.** Definition of congruent triangles
10. $AC = BC$	**10.** Definition of congruent segments
11. If a point lies on the perpendicular bisector of a segment, then it is equidistant from the endpoints	**11.** Law of Deduction

Now study the proof of the other half of the biconditional (the *if* part).

Given: $\overline{AB}$ and point P so that $AP = BP$

Prove: P lies on the perpendicular bisector of $\overline{AB}$

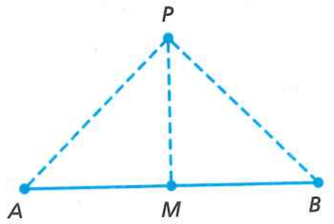


Figure 7.6

STATEMENTS	REASONS
1. $AP = BP$	**1.** Given
2. The midpoint M of $\overline{AB}$ exists	**2.** Completeness Postulate
3. $AM = MB$	**3.** Definition of midpoint
4. $\overline{AP} \cong \overline{BP}$; $\overline{AM} \cong \overline{BM}$	**4.** Definition of congruent segments
5. $\overline{MP} \cong \overline{MP}$	**5.** Reflexive property of congruent segments
6. $\triangle APM \cong \triangle BPM$	**6.** SSS
7. $\angle AMP \cong \angle BMP$	**7.** Definition of congruent triangles
8. $\angle AMP$ and $\angle BMP$ are right angles	**8.** Congruent angles that form a linear pair are right angles
9. $\overleftrightarrow{MP} \perp \overleftrightarrow{AB}$	**9.** Definition of perpendicular
10. $\overline{MP}$ bisects $\overline{AB}$	**10.** Definition of segment bisector (contains midpoint)
11. $\overleftrightarrow{MP}$ is the perpendicular bisector of $\overline{AB}$	**11.** Definition of perpendicular bisector
12. If P is equidistant from the endpoints of $\overline{AB}$, then it is on the perpendicular bisector of $\overline{AB}$	**12.** Law of Deduction

What happens if you draw the perpendicular bisectors of each side of a triangle? You should find that they are concurrent and that the distance from the point of concurrency to each vertex of the triangle is the same. Draw a circle through the vertices with the point of concurrency as the center. Because this circle circumscribes the triangle, the point of concurrency is called the *circumcenter*.

Theorem 7.6

Circumcenter Theorem. **The perpendicular bisectors of the sides of any triangle are concurrent at the circumcenter, which is equidistant from each vertex of the triangle.**

Given: $\triangle XYZ$, where line a is the perpendicular bisector of $\overline{ZY}$, line b is the perpendicular bisector of $\overline{XY}$, and line c is the perpendicular bisector of $\overline{XZ}$

Prove: a, b, and c intersect at point P, and P is equidistant from each vertex

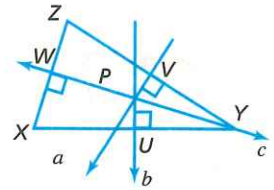


Figure 7.7

STATEMENTS	REASONS
1. a, b, and c are perpendicular bisectors to sides of $\triangle XYZ$	**1.** Given
2. a and b intersect at P	**2.** Definition of parallel (coplanar lines a and b are not parallel and so must intersect at some point)
3. $PZ = PY$; $PY = PX$	**3.** Any point on a perpendicular bisector of a segment is equidistant from the endpoints
4. $PZ = PX$	**4.** Transitive property of equality
5. P is on the perpendicular bisector, c, of $\overline{XZ}$	**5.** Any point equidistant from the endpoints of a segment is on the perpendicular bisector of the segment

Now try bisecting the angles of a triangle. The point of concurrency is the *incenter.*

Theorem 7.7

Incenter Theorem. **The angle bisectors of the angles of a triangle are concurrent at the incenter, which is equidistant from the sides of the triangle.**

You can also draw the altitude or the medians of triangles to obtain points of concurrency. The points of concurrency in these cases are called the *orthocenter* and the *centroid* respectively.

Definitions

An **altitude of a triangle** is a segment that extends from a vertex and is perpendicular to the opposite side.

A **median of a triangle** is a segment extending from a vertex to the midpoint of the opposite side.

Theorem 7.8

Orthocenter Theorem. **The lines that contain the three altitudes are concurrent at the orthocenter.**

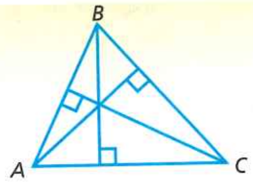
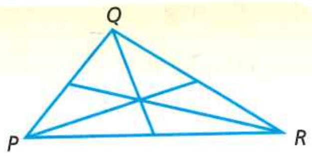


Figure 7.8

Theorem 7.9

Centroid Theorem. **The three medians of a triangle are concurrent at the centroid.**



The word *concurrent* comes from a word that means "to run together." This word applies in different but related senses both to lines and to Christians. According to Hebrews 12:1 and I Corinthians 9:24-26, each Christian life is hotly contested by the enemy and requires great exertion and struggle to the finish. Christians run this race concurrently as a team. Lines also run together, but they do so at the same point.

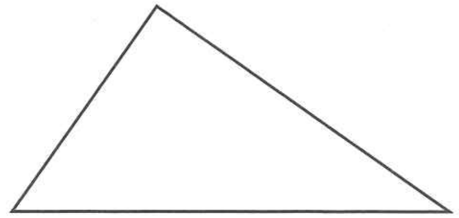
▶ A. Exercises

Draw four obtuse triangles. Use one triangle for each of the next four exercises.

1. Label the circumcenter C.
2. Label the incenter I.
3. Label the orthocenter O.
4. Label the centroid D.

Reproduce each triangle below four times, then find the centroid, incenter, orthocenter, and circumcenter.

5.



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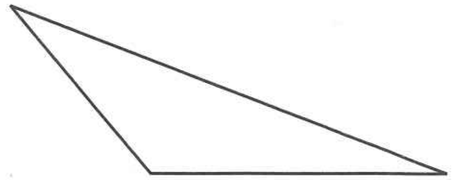
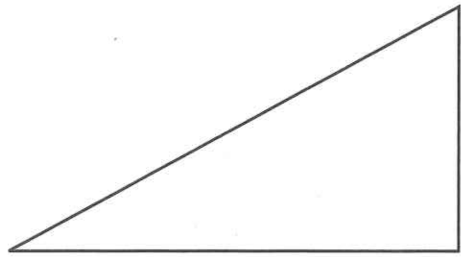
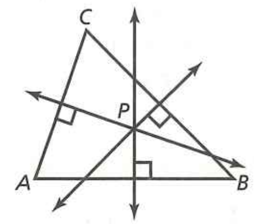


6.

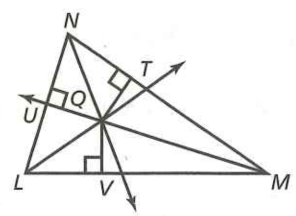


Use the diagram shown for exercises 8-10.



8. What is the name of point P? Draw a circle with a radius PA; what is the name of this circle?
9. If $PB = 10$ units, find PA.
10. If $PC = 15$ units, find PA.

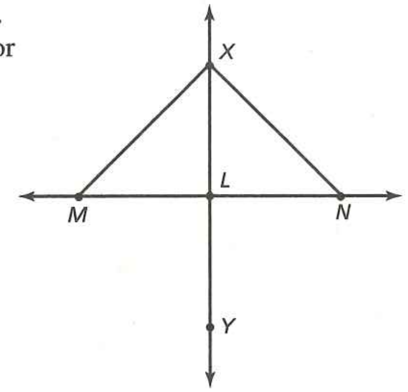
Use the diagram shown for exercises 11-13.



11. What is point Q called? Draw a circle with radius QV. What is the special name for this circle?
12. If $QV = 26$ units, find QT.
13. If $QT = 8$ units, find QU.

▶ B. Exercises

Use the diagram shown for exercises 14-15.



14. *Given:* $\overline{XY}$ is the perpendicular bisector of $\overline{MN}$ at point L

Prove: $\triangle MLX \cong \triangle NLX$

15. *Given:* $\overline{XL}$ is both a median and an altitude of $\triangle MXN$

Prove: $\triangle MLX \cong \triangle NLX$

16. *Given:* $\triangle ABC$ is equilateral; $\overline{BP}$ is a median

Prove: $\angle ABP \cong \angle CBP$

Prove the following theorems.

17. Any point on an angle bisector is equidistant from the sides of the angle. (*Hint:* Use congruent triangles.)

18. The median of an equilateral triangle is also an altitude.

▶ C. Exercises

19. For what kind of triangle are the incenter, orthocenter, centroid, and circumcenter the same? Prove your answer.

20. Prove that the medians from the base angles of an isosceles triangle are congruent.

▶ Dominion Thru Math

21. The trusses for the roof of a building are scalene right triangles sitting on the hypotenuses. The legs are 30 feet and 16 feet; overhangs on both sides are equal. How much overhang is on a 32-foot-wide building?

Cumulative Review

Given points A and B, consider the following:

$$\overline{AB},\ \emptyset,\ \overleftrightarrow{AB},\ AB,\ \overset{\circ}{\overrightarrow{AB}},\ \overrightarrow{AB}$$

22. Which symbol above is not a set?

23. Which set above is not a subset of any of the other sets?

24. Which set is a subset of all of the sets?

25. $\overleftrightarrow{AB}$ is a subset of which other sets above?

26. For which two of the sets is neither a subset of the other?

7.3 Exterior Angles and Inequalities

This room applies inequalities by incorporating rectangular windows of several unequal sizes.

The *exterior* refers to the outside of something. Indeed, "man looketh on the outward appearance, but the Lord looketh on the heart" (I Sam.16:7). While only God can judge men justly, we are to evaluate professions of salvation by their exterior fruit (Matt. 7:15-20).

The *exterior angle* of a triangle is an angle on the outside. $\angle ABD$ is an exterior angle of $\triangle BCD$ in figure 7.9. Draw the other five exterior angles of the triangle by extending different sides. Exterior angles have some special properties.

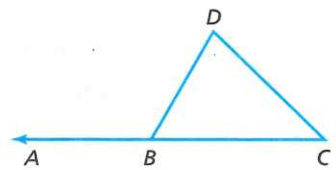


Figure 7.9

Definitions

An **exterior angle** of a triangle is an angle that forms a linear pair with one of the angles of the triangle.

The **remote interior angles** of an exterior angle are the two angles of the triangle that do not form a linear pair with a given exterior angle.

In figure 7.9, the remote interior angles of $\angle ABD$ are $\angle D$ and $\angle C$. Theorem 7.10 shows a relationship between an exterior angle and its two remote interior angles.

Theorem 7.10

Exterior Angle Theorem. **The measure of an exterior angle of a triangle is equal to the sum of the measures of its two remote interior angles.**

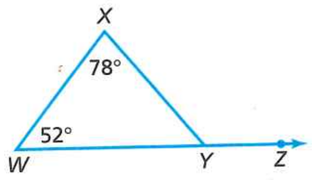


Figure 7.10

For example, the measure of $\angle XYZ$ in figure 7.10 is 130 degrees. Do you understand why? It makes sense when you realize that $m\angle WYX$ is part of two 180-degree totals. The triangle and the linear pair suggest a proof for the theorem.

Given: $\triangle HIJ$ with exterior $\angle JHG$

Prove: $m\angle JHG = m\angle I + m\angle J$

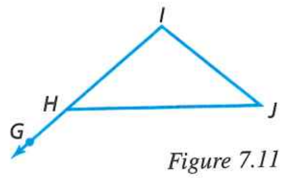


Figure 7.11

STATEMENTS	REASONS
1. $\triangle HIJ$ with exterior $\angle JHG$	1. Given
2. $\angle IHJ$ and $\angle JHG$ form a linear pair	2. Definition of exterior angle
3. $\angle IHJ$ and $\angle JHG$ are supplementary	3. Linear pairs are supplementary
4. $m\angle IHJ + m\angle JHG = 180$	4. Definition of supplementary
5. $m\angle IHJ + m\angle I + m\angle J = 180$	5. The measures of the angles of a triangle total 180
6. $m\angle IHJ + m\angle JHG = m\angle IHJ + m\angle I + m\angle J$	6. Transitive property of equality
7. $m\angle JHG = m\angle I + m\angle J$	7. Addition property of equality

The exterior angle theorem also shows that the exterior angle is larger than either remote interior angle. The proof of the inequality "larger than" will require properties of inequality. Since the rest of this chapter involves inequalities, review the properties of inequality closely. Assume a, b, and c are real numbers.

Inequality Facts	
Addition property	If $a < b$, then $a + c < b + c$
Multiplication property	If $a < b$ and $c > 0$, then $ac < bc$
	If $a < b$ and $c < 0$, then $ac > bc$
Transitive property	If $a < b$ and $b < c$, then $a < c$
Definition of greater than	If $a = b + c$ and $c > 0$, then $a > b$

Theorem 7.11

Exterior Angle Inequality. **The measure of an exterior angle of a triangle is greater than the measure of either remote interior angle.**

Given: $\triangle ABC$ and exterior $\angle DAB$

Prove: $m\angle DAB > m\angle B$ and $m\angle DAB > m\angle C$

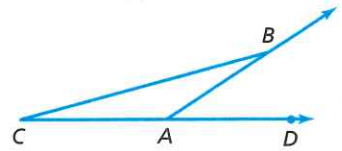


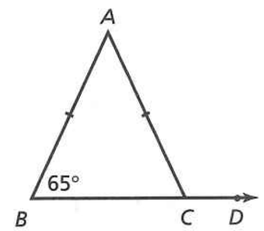
Figure 7.12

STATEMENTS	REASONS
1. $\triangle ABC$ and exterior angle $\angle DAB$	1. Given
2. $m\angle DAB = m\angle B + m\angle C$	2. Exterior Angle Theorem
3. $m\angle DAB > m\angle B$	3. Definition of greater than
4. $m\angle DAB > m\angle C$	4. Definition of greater than

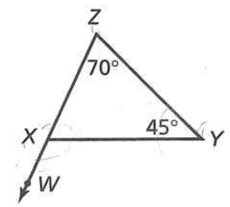
▶ A. Exercises

Find the measures of the indicated angles.

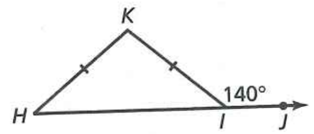
1. $m\angle A$
   $m\angle BCA$
   $m\angle ACD$



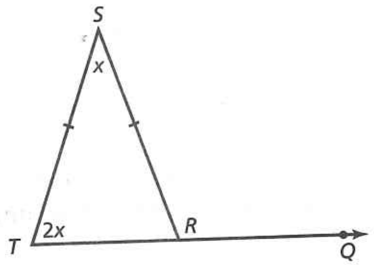
2. $m\angle ZXY$
   $m\angle WXY$



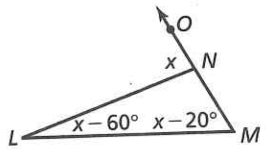
3. $m\angle KIH$
$m\angle H$
$m\angle K$



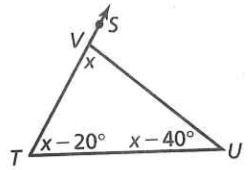
4. $m\angle S$
$m\angle T$
$m\angle TRS$
$m\angle QRS$



5. $m\angle ONL$
$m\angle LNM$
$m\angle L$
$m\angle M$

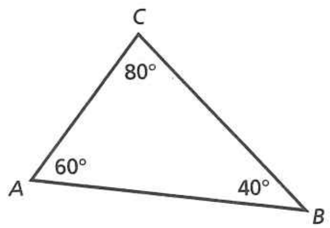


6. $m\angle SVU$
$m\angle UVT$
$m\angle T$
$m\angle U$



Refer to $\triangle ABC$; give the measure of the exterior angles at each vertex.

7. A
8. B
9. C



10. Do the largest interior and exterior angles of the triangle occur at the same vertex?

Answer each question.

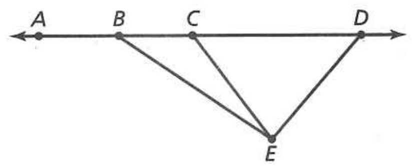
11. If $\triangle ABC$ is equilateral, what is the measure of an exterior angle?
12. What are the measures of the acute angles in an isosceles right triangle?

▶ B. Exercises

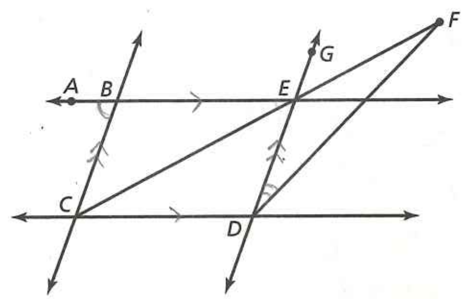
13. If an exterior angle at the vertex of an isosceles triangle measures 78 degrees, what are the measures of the base angles?
14. What is the measure of an angle determined by a side and a diagonal of a square?
15. If $ABCD$ is a rhombus and C-D-E, what is the relationship between $m\angle EDA$ and $m\angle DAC$?

Use theorems on exterior angles to prove exercises 16-20.

16. Angles adjacent to the hypotenuse of a right triangle are complementary.
17. A right triangle has two acute angles.
18. The exterior angle at the vertex of an isosceles triangle measures twice the measure of one of the base angles.
19. Prove that $m\angle ABE > m\angle CED$.



20. *Prove:* If $BCDE$ is a parallelogram, then $m\angle ABC > m\angle FDE$.



▶ C. Exercises

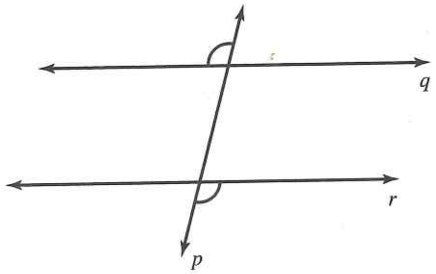
Exercises 21-23 develop the sum of the measures of the exterior angles for any convex polygon. Use only one exterior angle at each vertex. Find the sum of the measures for each polygon and justify your answer.

21. triangle
22. convex quadrilateral
23. n-gon

Cumulative Review

How does each diagram illustrate a method of proving two lines parallel? Identify the parallel lines in each case.

24.



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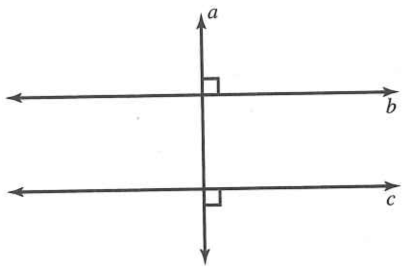

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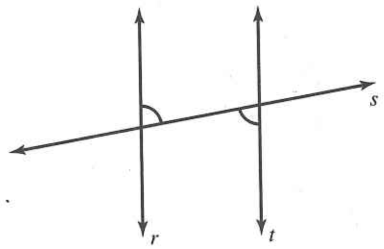

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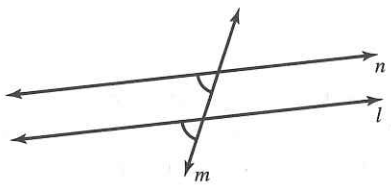

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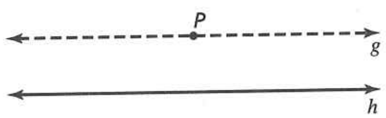


GEOMETRY AROUND US

GEOMETRY AND CARPENTRY

Miter saws, planes, and air nailers are tools for cutting, working, and joining timbers. A carpenter depends on these tools as well as others of his trade. He uses a framing square to test right angles between adjacent surfaces. He uses a bevel to check acute and obtuse angles. A gauge enables him to mark lines parallel to a surface.

Did you know that geometry is another essential tool of the carpenter? A carpenter faces a number of geometric problems, from laying out the foundation of a structure to cutting the proper angles for the rafters that support the roof. The foundation for a house usually consists of one or more rectangles that must be laid out perfectly. A carpenter cannot simply measure the lengths of the sides of the quadrilateral because that would guarantee only a parallelogram. Using theorems from geometry, he can either check that one of the angles is a right angle, or he can measure the lengths of the diagonals to see if they are the same.

A carpenter must also be able to read an architect's drawing. This is essential if he is going to build a house exactly the way the future owner wants it. If he makes

A carpenter nails boards for the roof with a pneumatic (air-driven) tool.

a utility room too small for the washer and dryer, it may be expensive to correct the mistake. If a door is too small or the stairs too narrow, it may be impossible to bring in the furniture. Reading these drawings requires a knowledge of proportion and ratio. The carpenter must also recognize errors in the drawings, if there are any.

Carpenters frame trapezoids, rectangles, and an octagon for windows as they construct this house.

When making door and window frames, the carpenter must be able to measure and construct right angles. When erecting the interior walls and their supports, he deals with perpendicular and parallel lines. He applies the theorem that in a plane, if two lines are perpendicular to the same line, then they are parallel. If an arch is needed, a carpenter must know whether it is to be a semicircle, pointed arch, parabola, or catenary curve and apply principles for building it.

Another geometric task that the carpenter must handle is constructing a flight of stairs. When handling such a job, he uses, perhaps without knowing its official name, the Pythagorean theorem. He uses a carpenter's square—a device that looks like a right triangle without its hypotenuse—for measuring and for marking each stair. After determining how many feet the stairs will rise vertically, he can divide by the maximum step height (riser) allowed by the local building code and determine the number of steps required.

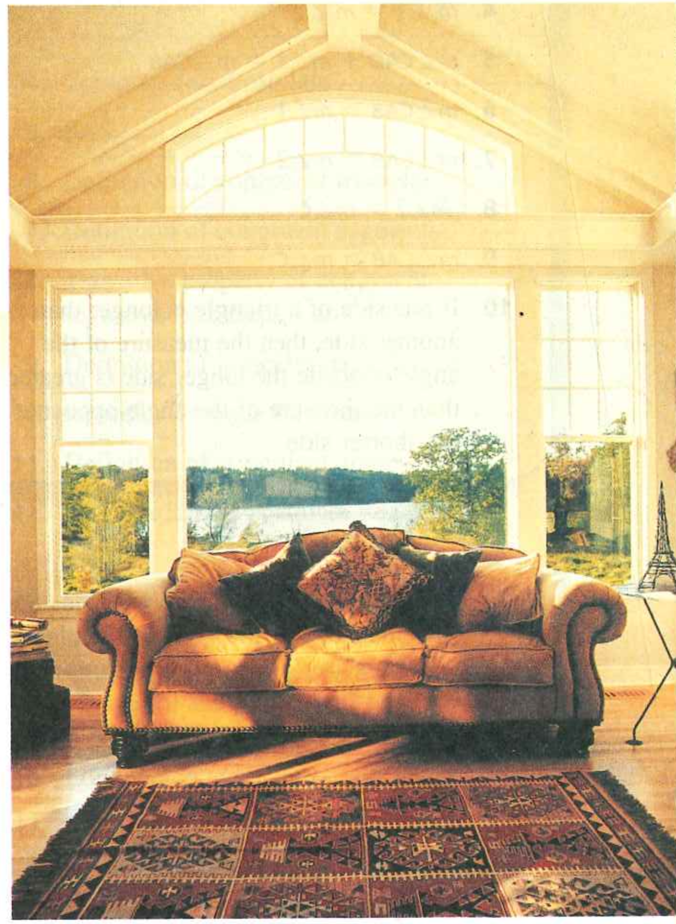
When handling such a job, he uses, perhaps without knowing its official name, the Pythagorean theorem.

Building codes also specify the minimum width of tread (where you place your foot). Multiplying this tread size by the number of steps, the carpenter can determine the horizontal distance required for the staircase. He then must determine a set ratio for each individual stair and use his carpenter's square to check each step.

Anyone who builds something, whether a house or a bookcase, uses the skills of the carpenter. The principles will serve you well as you apply them in the shop. Geometry is indeed one of a carpenter's most essential tools.

7.4 Inequalities

The Exterior Angle Inequality is not the only important inequality for triangles. You will learn two more in this section. The first is a biconditional statement, but only one part will be proved.



Scalene right triangles flank the arched window of this home in Stillwater, Minnesota.

Theorem 7.12
***Longer Side Inequality.* One side of a triangle is longer than another side if and only if the measure of the angle opposite the longer side is greater than the measure of the angle opposite the shorter side.**

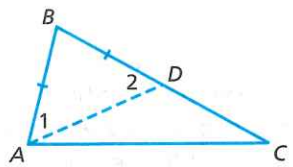


Figure 7.13

We will prove the *if* part of the Longer Side Inequality.

If $BC > AB$, then $m\angle A > m\angle C$.

Given: $\triangle ABC$; $BC > AB$

Construct: $\overline{AD}$ such that B-D-C and $\overline{AB} \cong \overline{BD}$ (auxiliary line by Completeness Postulate)

Prove: $m\angle A > m\angle C$

STATEMENTS	REASONS
1. $\triangle ABC$; $BC > AB$; $\overline{AB} \cong \overline{BD}$	1. Given
2. $\triangle ABD$ is an isosceles triangle	2. Definition of isosceles triangle
3. $\angle 1 \cong \angle 2$	3. Isosceles Triangle Theorem
4. $m\angle 1 = m\angle 2$	4. Definition of congruent angles
5. $m\angle CAD + m\angle 1 = m\angle CAB$	5. Angle Addition Postulate
6. $m\angle CAB > m\angle 1$	6. Definition of greater than
7. $m\angle CAB > m\angle 2$	7. Substitution (step 4 into 6)
8. $m\angle 2 > m\angle C$	8. Exterior Angle Inequality
9. $m\angle CAB > m\angle C$	9. Transitive property of inequality
10. If one side of a triangle is longer than another side, then the measure of the angle opposite the longer side is greater than the measure of the angle opposite the shorter side	10. Law of Deduction

Theorem 7.13

Hinge Theorem. **Two triangles have two pairs of congruent sides. If the measure of the included angle of the first triangle is larger than the measure of the other included angle, then the opposite (third) side of the first triangle is longer than the opposite side of the second triangle.**

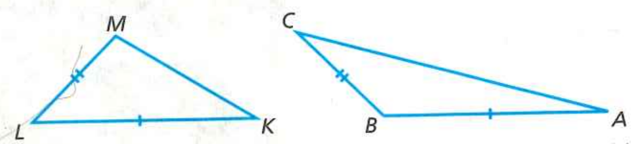


Figure 7.14

Given: $\triangle KLM$ and $\triangle ABC$ with $AB \cong KL$; $BC \cong LM$; and $m\angle ABC > m\angle KLM$

Draw: $\overrightarrow{BD}$ so that $\triangle ABD \cong \triangle KLM$

Prove: $AC > KM$

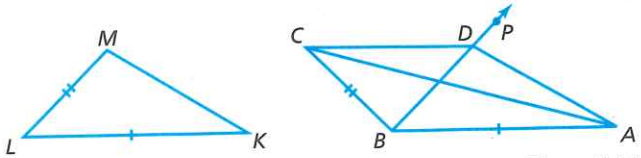


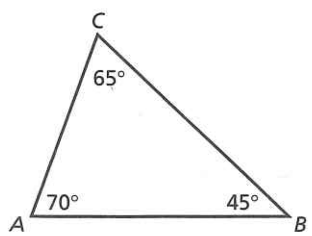
Figure 7.15

STATEMENTS	REASONS
1. $\triangle ABC$; $\triangle KLM$; $\overline{AB} \cong \overline{KL}$; $\overline{BC} \cong \overline{LM}$ and $m\angle ABC > m\angle KLM$	1. Given
2. $m\angle ABC = d + m\angle KLM$ $(d > 0)$	2. Definition of greater than
3. Draw $\overrightarrow{BP}$ with P on the same side of $\overleftrightarrow{AB}$ as C, so that $\angle ABP \cong \angle KLM$	3. Auxiliary line (Continuity Postulate)
4. Mark D on $\overrightarrow{BP}$ so that $\overline{BD} \cong \overline{LM}$	4. Completeness Postulate
5. $\triangle ABD \cong \triangle KLM$	5. SAS
6. $\overline{AD} \cong \overline{KM}$	6. Definition of congruent triangles
7. $AD = KM$	7. Definition of congruent segments
8. $\overline{BC} \cong \overline{BD}$	8. Transitive property of congruent segments (see steps 1 and 4)
9. $\triangle BCD$ is isosceles	9. Definition of isosceles triangle
10. $\angle BCD \cong \angle BDC$	10. Isosceles Triangle Theorem
11. $m\angle BCD = m\angle BDC$	11. Definition of congruent angles
12. $m\angle ADC = m\angle ADB + m\angle BDC$; $m\angle BCD = m\angle BCA + m\angle DCA$	12. Angle Addition Postulate
13. $m\angle ADC > m\angle BDC$; $m\angle BCD > m\angle DCA$	13. Definition of greater than
14. $m\angle ADC > m\angle BCD$	14. Substitution (step 11 into 13)
15. $m\angle ADC > m\angle DCA$	15. Transitive property of inequality
16. $AC > AD$	16. Longer side inequality
17. $AC > KM$	17. Substitution (step 7 into 16)

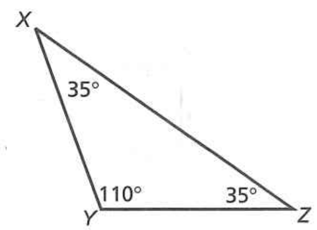
▶ A. Exercises

Give the order of sides from smallest to largest.

1.



2.



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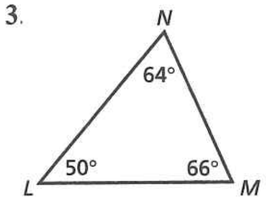

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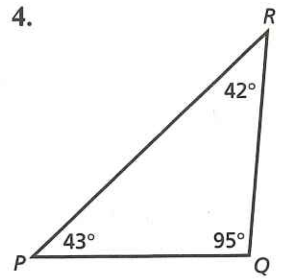
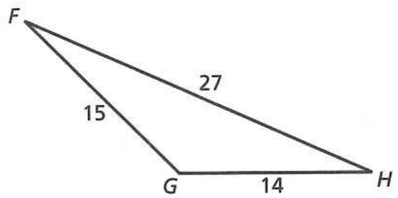


Give the order of angles from smallest to largest.

5.


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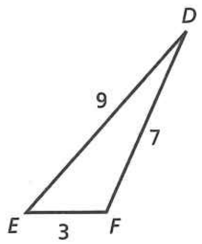

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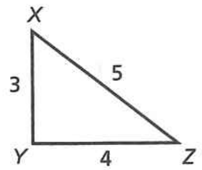
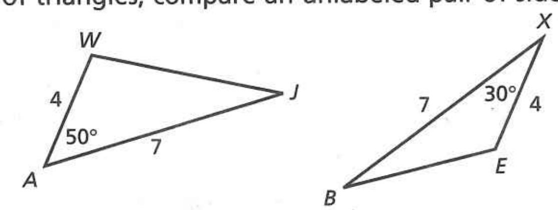


For each pair of triangles, compare an unlabeled pair of sides or angles.

8.


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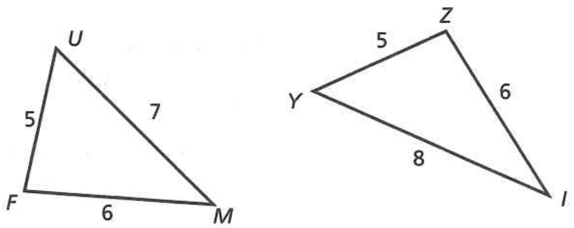
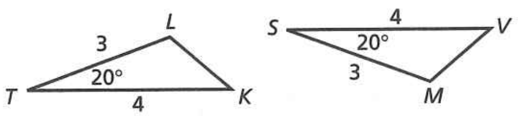
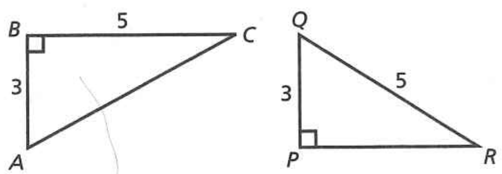


10.


11.


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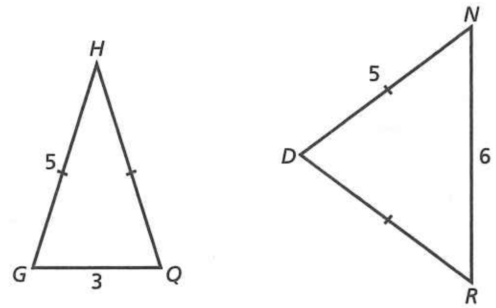


Prove the following statement.

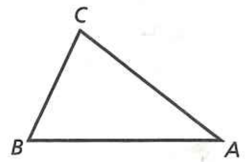
13. In a right triangle the hypotenuse is the longest side.

▶ B. Exercises

Prove the following statement.

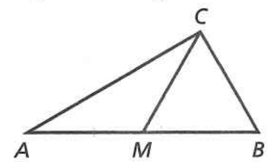
14. The shortest segment from a point to a line is a perpendicular segment.

15. *Given:* $AB > AC$; $AC > BC$
Prove: $m\angle C > m\angle A$



Prove exercises 16-17 using the following diagram and given information.

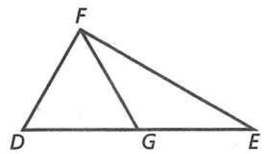
Given: $\overline{MB}$ is the base of isosceles $\triangle BCM$ and M is the midpoint of $\overline{AB}$



16. *Prove:* $AC > CM$

17. *Prove:* $m\angle B > m\angle A$

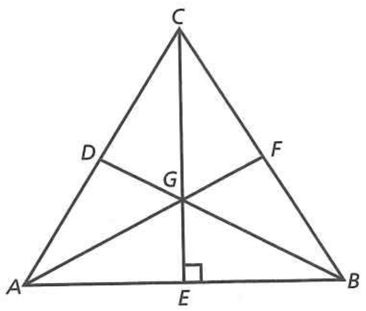
18. *Given:* $\overline{FG}$ is a median;
$m\angle FGE > m\angle FGD$
Prove: $m\angle D > m\angle E$



19. The longest side of an obtuse triangle is opposite the obtuse angle.

▶ C. Exercises

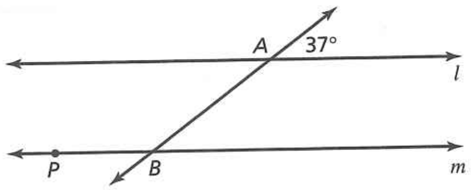
20. Prove that the centroid of an equilateral triangle divides the medians into segments of unequal lengths. Write your proof in paragraph form stating only the key steps.



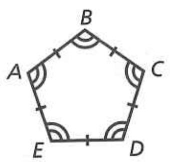
Cumulative Review

Give an algebraic expression or a numerical value for each indicated angle.

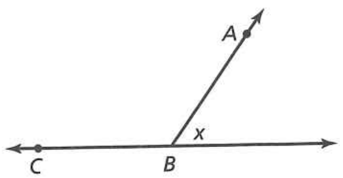
21. $m\angle PBA$ if $l \parallel m$



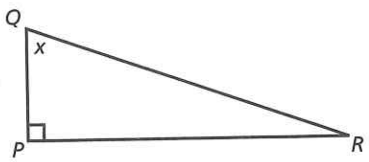
22. $m\angle A$



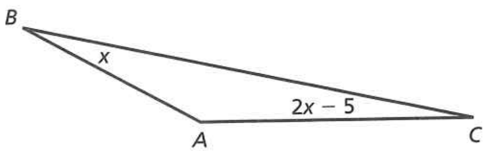
23. $m\angle ABC$



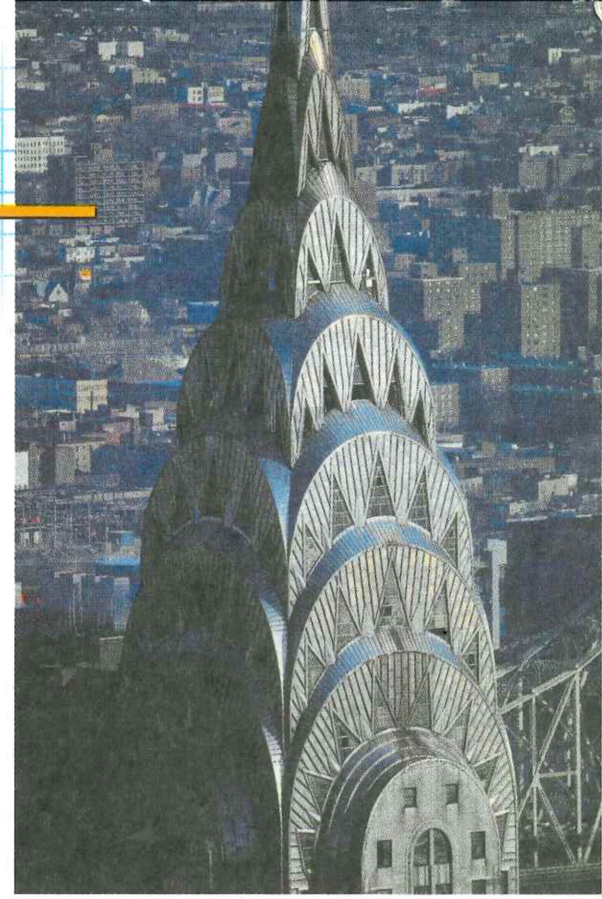
24. $m\angle R$



25. $m\angle A$



7.5 Triangle Inequality



The Chrysler Building in New York City incorporates triangles on its spire; it was the tallest skyscraper in the world for several months before the completion of the Empire State Building.

In each of the three diagrams in figure 7.16, $BC = 8$. In diagram (*a*), $AB + AC = 8$. According to the definition of betweenness, the points are collinear and no triangle is formed.

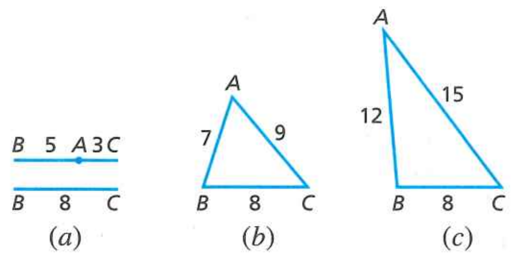


Figure 7.16

In diagrams (*b*) and (*c*), $AB + AC > 8$, and triangles are formed. This illustration suggests the Triangle Inequality Theorem.

Theorem 7.14

***Triangle Inequality.* The sum of the lengths of any two sides of a triangle is greater than the length of the third side.**

Given: $\triangle XYZ$ with no side longer than $\overline{XZ}$

Auxiliary line: $\overleftrightarrow{PY} \perp \overleftrightarrow{XZ}$ at point P

Prove: $XZ + YZ > XY$
$XZ + XY > YZ$
$XY + YZ > XZ$

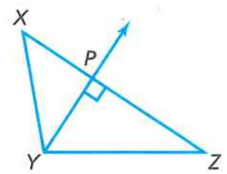


Figure 7.17

Trichotomy guarantees that you can list the three sides of a triangle in order of length from shortest to longest. (If two are the same length, either order will do.) Label side XZ so that there is no longer side. Since $XZ \geq XY$, it follows that $XZ + YZ > XY$ (distance $YZ > 0$). Similarly, $XZ \geq YZ$, so $XZ + XY > YZ$. These two inequalities were easy to prove, but the third is harder and requires the altitude from Y to $\overline{XZ}$ as an auxiliary line.

STATEMENTS	REASONS
1. $\triangle XYZ$ with $\overline{XZ}$ the longest side, altitude $\overline{YP}$	1. Given
2. $\overline{YP} \perp \overline{XZ}$	2. Definition of altitude
3. $\angle XPY$ and $\angle ZPY$ are right angles	3. Definition of perpendicular
4. $\triangle XPY$ and $\triangle ZPY$ are right triangles	4. Definition of right triangle
5. $XY > XP$, $YZ > PZ$	5. Hypotenuse is longest side
6. $XY + YZ > XP + PZ$	6. Addition property of inequality
7. $XP + PZ = XZ$	7. Definition of betweenness
8. $XY + YZ > XZ$	8. Substitution (step 7 into 6)

Now you can see at a glance that no triangle could exist with sides of lengths 2, 3, and 7. The diagram shows that the short sides are too short to intersect. The triangle inequality requires that $2 + 3 > 7$. Since this is false, the triangle inequality explains why no such triangle exists.

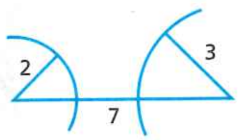


Figure 7.18

This principle applies in many practical situations as illustrated in the example.

EXAMPLE In a business office the equipment must be arranged carefully for efficient use. Becky uses primarily the computer on her desk, the photocopier, and the fax machine. A triangular arrangement is desired to facilitate access between all three machines. The distance between her desk and the photocopier must be the shortest, and the total distance between the three items should not exceed 20 feet. The following table shows distances between the pieces of equipment. Which of these is possible and which is impossible? Why?

	Computer-Copier	Copier-Fax	Computer-Fax
1.	5 feet	7 feet	6 feet
2.	3 feet	6 feet	9 feet

Continued ▶

Answer *Layout 1* has a total distance between items of 18 feet, and the computer-copier distance is the shortest as desired. Every combination of two sides is greater than the third side, so layout 1 forms a legitimate triangular layout.

Layout 2 has a total distance of 18 feet, with the computer-copier distance being the shortest. However, the sum of the first two sides equals the third side. This combination is not triangular and therefore not efficient. Notice that the three items are collinear, so the photocopier is in the way of moving from the computer to the fax machine.

▶ A. Exercises

State whether triangles with the following side lengths exist. If not, state the reason.

	a	*b*	*c*	
1.	5	9	10	
2.	2	7	11	
3.	8	4	12	
4.	4	7	8	
5.	6	3	3	

AB, *BC*, and *AC* are given. Make a sketch and identify the type of triangle. If none is formed, explain why (impossible, collinear).

6. $AB = 4$, $BC = 7$, $AC = 5$
7. $AB = 3$, $BC = 5$, $AC = 2$
8. $AB = 1$, $BC = 1$, $AC = 1$
9. $AB = 5$, $BC = 3$, $AC = 1$
10. $AB = 2$, $BC = 5$, $AC = 2$
11. $AB = 5$, $BC = 3$, $AC = 3$
12. $AB = 5$, $BC = 5$, $AC = 10$

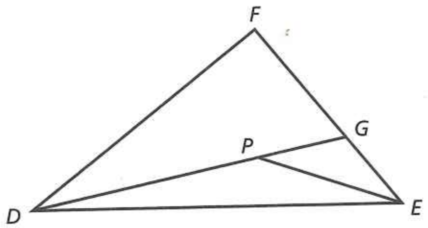
▶ B. Exercises

Refer to Becky's office in the example in this section. Are the following measurements possible? If not, explain why.

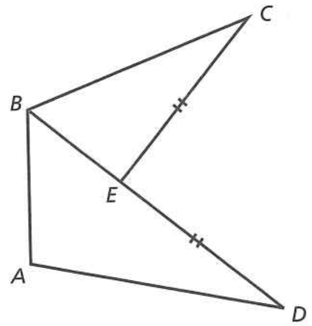
	Computer-Copier	Copier-Fax	Computer-Fax	
13.	5 feet	7 feet	8 feet	
14.	9 feet	4 feet	6 feet	
15.	6 feet	12 feet	10 feet	
16.	4 feet	10 feet	5 feet	
17.	5 feet	9 feet	6 feet	

18. Explain why the sum of the lengths of the legs of any right triangle is always greater than the length of the hypotenuse.

19. *Given:* $\triangle DEF$ with point P in the interior
Prove: $DF + FE > DP + PE$

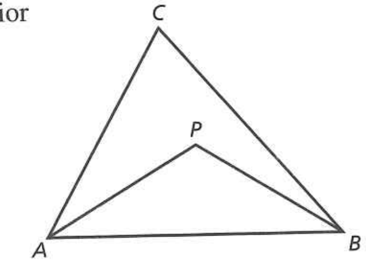


20. *Given:* $\overline{EC} \cong \overline{ED}$
Prove: $AB + AD > BC$



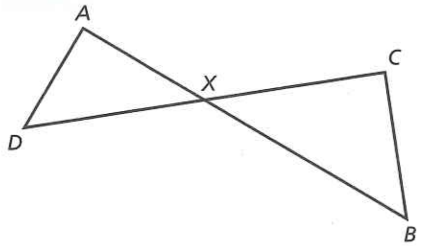
▶ C. Exercises

21. *Given:* $\triangle ABC$ with point P in the interior
Prove: $m\angle APB > m\angle ACB$



Cumulative Review

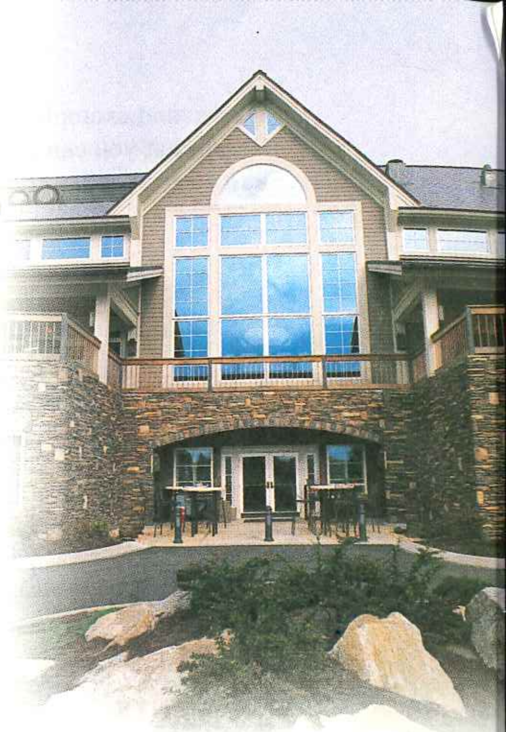
Supply the missing reasons needed to complete the following proof.
Given: $\overleftrightarrow{AD} \perp \overleftrightarrow{AB}$, $\overleftrightarrow{CB} \perp \overleftrightarrow{CD}$
Prove: $\angle D \cong \angle B$



STATEMENTS	REASONS
22. $\overleftrightarrow{AD} \perp \overleftrightarrow{AB}$; $\overleftrightarrow{CB} \perp \overleftrightarrow{CD}$	22.
23. $\angle A$ and $\angle C$ are right angles	23.
24. $\angle A \cong \angle C$	24.
25. $\angle AXD \cong \angle CXB$	25.
26. $\angle D \cong \angle B$	26.

7.6 Parallelograms

Remember that a parallelogram is a quadrilateral in which both pairs of opposite sides are parallel. Study the examples of proofs involving parallelograms. These important theorems will enable you to prove many other theorems.



The rectangles and squares of these windows are special parallelograms.

Theorem 7.15
The opposite sides of a parallelogram are congruent.

Given: Parallelogram $ABCD$
Prove: $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{AD}$
Draw auxiliary diagonal $\overleftrightarrow{BD}$.

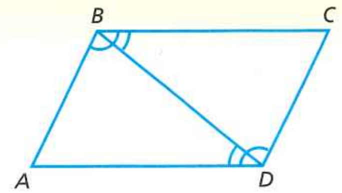


Figure 7.19

STATEMENTS	REASONS
1. $ABCD$ is a parallelogram	1. Given
2. $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$; $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$	2. Definition of parallelogram
3. Draw $\overleftrightarrow{BD}$	3. Line Postulate
4. $\angle ABD \cong \angle CDB$; $\angle CBD \cong \angle ADB$	4. Parallel Postulate
5. $\overline{BD} \cong \overline{BD}$	5. Reflexive property of congruent segments
6. $\triangle ABD \cong \triangle CDB$	6. ASA
7. $\overline{AB} \cong \overline{CD}$; $\overline{AD} \cong \overline{BC}$	7. Definition of congruent triangles